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A yield-resilient 1 Mb RRAM macro with bipolar forming and cell-by-cell write verification

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ABSTRACT

Resistive RAM(RRAM) has emerged as a promising candidate for the next generation non-volatile memories (NVMs) due to its low write voltage, compact cell area, and CMOS compatibility. In this work, a 1 Mb RRAM macro with bipolar forming is proposed to reduce the forming voltage and improve programming yield. In addition, a cell-by-cell write-verification scheme is introduced to improve the bit-level programming yield by verifying the switching result of each programmed cell. The test chip, fabricated using 40-nm CMOS technology, achieves a bit-level programming yield of over 90% and demonstrates a 7–10% improvement in forming yield compared with the unipolar forming case under the same bias conditions, while occupying a core area of 2.34 mm^2. Furthermore, the measured results establish a 90% programming-yield criterion for selecting the write-voltage condition, providing a practical guideline for yield-robust RRAM programming.

1. Introduction

Flash memory has been the leading non-volatile memory (NVM) technology due to its high-density data storage capabilities, maintaining its dominance at least through the 2X-nm generation. [1,2]. However, the intrinsic physical characteristics of flash memory faces significant limitations beyond certain technical nodes, such as high leakage current and slow write speed. As a result, novel NVM technologies are being developed to achieve high density, low power, and fast operation. Several emerging NVMs, such as magnetoresistive random-access memory (MRAM) [3,4], ferroelectric RAM (FERAM) [5], phase-change RAM (PCRAM) [6,7], and resistive RAM (RRAM) [8–11,12] are introduced as promising alternatives due to improved write speed and reduced power consumption compared to embedded flash memory. However, MRAM has limited density, FeRAM faces challenges in scalability, and PCRAM suffers from high energy consumption during write operations. RRAM is considered an excellent NVM among these emerging NVM technologies due to its advantageous characteristics, such as low operating voltage (<3V), fast switching speed (<10ns), compact area (<10 nm), large resistance ratio(10~1000), and excellent compatibility with CMOS technology [2,13]. These advantages of RRAM make it appealing for next-generation NVM memory in high-density and high-speed NVM applications. The 1T1R logic process-compatible RRAMs in Refs. [14–18] demonstrated strong suitability for embedded NVM applications that operate at low power and require fast read, particularly for energy-constrained devices. Our prior work in Ref. [19] demonstrated a low-power 1 Mb RRAM macro with reliable sensing by a two-stage voltage-mode sense amplifier (VSA) and improved read speed. In this work, we proposed an improved 1 Mb RRAM macro that employs dual-mode write drivers to support unipolar and bipolar forming. A write verification scheme is also employed to improve the write yield of the developed RRAM.

2. Background

2.1. 1T1R RRAM bitcell and basic operations

RRAM is a metal-insulator-metal (MIM) structure that can switch between a high-resistance state (HRS) and a low-resistance state (LRS), depending on the direction of the current flowing through it. The 1T1R bitcell consists of an NMOS transistor and an RRAM connected in series, as shown in Fig. 1. In this configuration, the access transistor selects and

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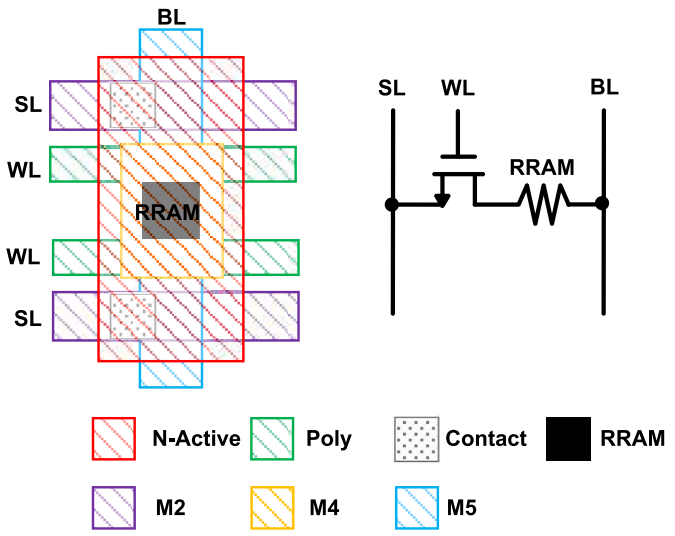


Fig. 1. RRAM bitcell layout (left) and schematic (right).

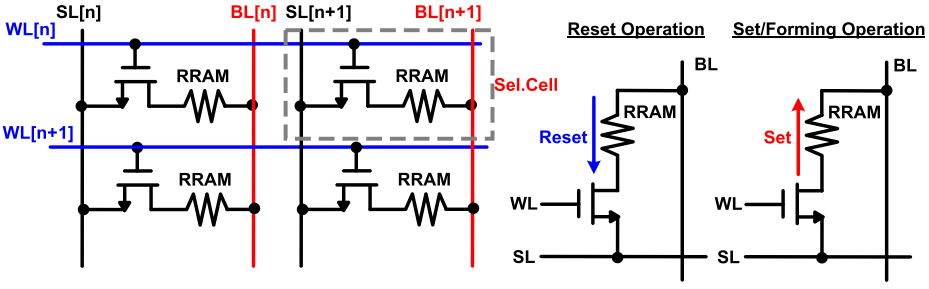
controls the RRAM by enabling the word-line (WL), while the RRAM functions as a storage device by switching the state between HRS ("0") and LRS ("1"). To enable column-by-column programming access and cell-by-cell write verification, a split gate and source-line (SL) design is implemented, resulting in a bitcell area of 0.135 μm^2. Fig. 2 illustrates the fundamental operations performed on the RRAM array. Initially, fresh cells require a once-forming operation to induce soft breakdown of the dielectric by applying high forming voltages ($V_{WL\text{-}FOM}$ and $V_{SL\text{-}FOM}$) to the WL and the SL, while grounding the BL before programming. It is a one-time process to switch the RRAM devices from initial HRS to LRS. However, unselected cells in the same row have 0 V on both BL and SL. This ensures that there is no voltage difference across the unselected cells, which prevents unwanted forming operations. After the forming operation, RRAM cells are ready for programming. "SET" refers to the RRAM cell transition from HRS to LRS, corresponding to writing "1." It is similar to the forming operation. However, the applied bias voltage is lower than the forming voltage. The opposite process, known as "RESET", switches the selected RRAM cell from LRS to HRS by applying a write voltage to the BL. Simultaneously, the BLs and SLs of unselected cells are grounded, as shown in the table of Fig. 2. During the read operation, the BL is pre-charged with a low voltage to avoid unwanted partial resetting of the RRAM and state flipping.

2.2. Motivation for realizing bipolar forming operation

In RRAM operations, forming is an essential process for creating conductive filaments to activate resistive switching. Typically, the forming voltage is higher than the SET and RESET voltage, which switches the device from HRS to LRS or from LRS to HRS, respectively. As the device size decreases, reducing the forming voltage becomes a critical challenge. Unipolar forming has been widely used as a mainstream forming method by applying a single pulse to one terminal of an RRAM device. Previous studies have mainly focused on reducing the forming voltage and improving programming behavior, since device-to-device variation in RRAM cells and the reduced voltage tolerance of scaled access transistors are critical issues for practical RRAM applications. As shown in Fig. 3, bipolar forming, which applies opposite-polarity pulses to both terminals of an RRAM device, has been employed to reduce the forming voltage. The reduced forming voltage under bipolar forming can be understood from the conductive-filament formation mechanism in oxide-based RRAM. During forming, oxygen-vacancy generation and oxygen-ion migration create a conductive path in the switching layer. Compared with unipolar forming, bipolar forming applies opposite-polarity bias to the BL and SL terminals, which can assist the electric-field-driven redistribution of oxygen vacancies and oxygen ions. Therefore, conductive-filament formation can be initiated at a lower externally applied forming voltage, which is consistent with prior bipolar-forming studies [20,21,22].

2.3. Motivation of write verification concept

During the write operation, reliable resistance switching is critical for practical RRAM applications. Conventionally, a fixed voltage amplitude and pulse width are used for programming. However, due to the stochastic formation and rupture of conductive filaments, the switching behavior of RRAM cells exhibits significant variability. Fast-switching cells may suffer from excessive energy consumption, long stress time, endurance degradation, and over-write issues, as depicted in Fig. 4(a) [23,24]. In contrast, slow-switching cells may not be fully programmed under a fixed voltage amplitude or pulse width, leading to



	WL		SL		BL	
	Sel. Cell	Unselected Cell	Sel. Cell	Unselected Cell	Sel. Cell	Unselected Cell
FORM	$V_{WL\text{-}FOM}$	0V	$V_{SL\text{-}FOM}$	0V	0V	0V
SET	$V_{WL\text{-}SET}$	0V	$V_{SL\text{-}SET}$	0V	0V	0V
RESET	$V_{WL\text{-}RST}$	0V	0V	0V	$V_{BL\text{-}RST}$	0V
READ	$V_{WL\text{-}RD}$	0V	0V	0V	$V_{BL\text{-}RD}$	0V

Fig. 2. RRAM array structure and basic operations.

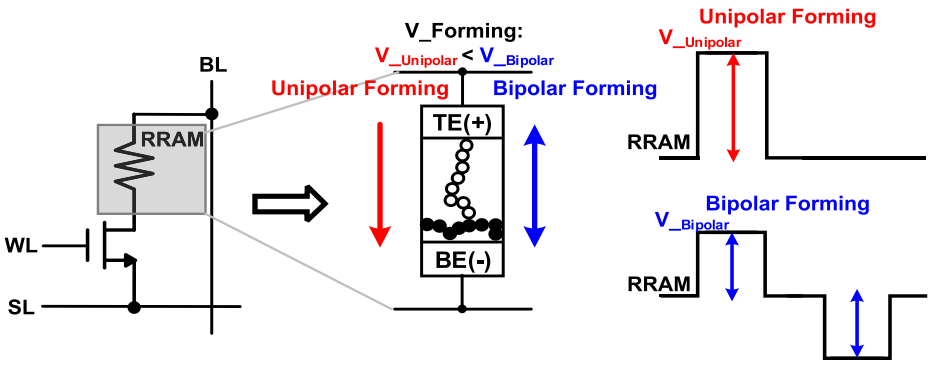


Fig. 3. Simplified operation of unipolar forming and bipolar forming.

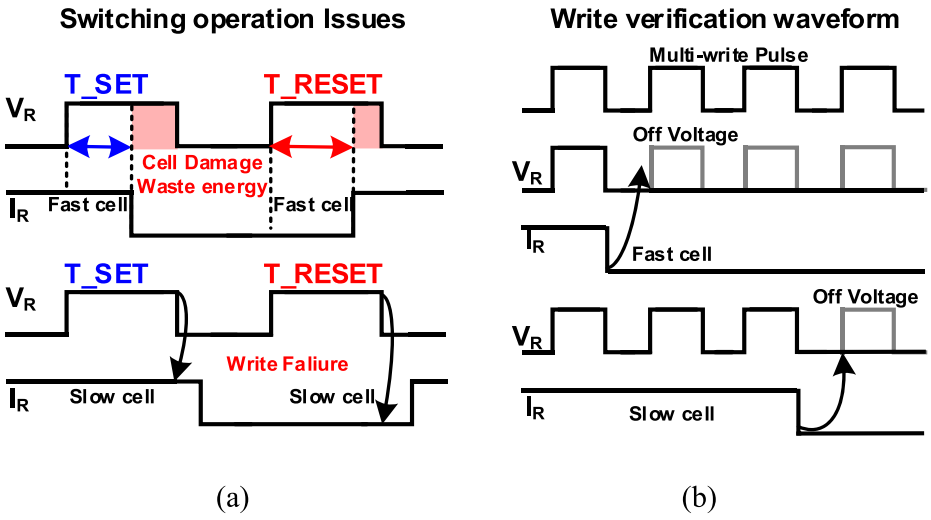


Fig. 4. Waveform of (a) switching issues and of (b) write verification operation.

SET or RESET failures. These variations reduce the R-ratio (R_{HRS} /R_{LRS}) and can affect retention and endurance characteristics [23–27].

Prior write-verification schemes have mainly focused on either verifying whether the cell conductance falls within a target range for multilevel operation or terminating the write operation when the cell resistance reaches a predefined threshold. For example, Shim et al. used an iterative two-step write–verify scheme to place multilevel RRAM cell conductance within a target range [23], while Chen et al. used a self-write termination scheme to suppress tail bits and avoid over-write during SET/RESET operations [28]. In addition, prior RRAM macro studies have reported that device-to-device and cycle-to-cycle switching variability, including set/reset-time variation and resistance variation, can degrade programming yield, read margin, and resistance distribution [8], [24]. When a fixed or repeated programming pulse is applied even after the target resistance state has already been reached, unnecessary write-bias application, write disturbance, or over-set/over-reset effects can occur [8], [24].

Motivated by these issues, this work adopts a cell-by-cell write-verification scheme to improve the sampled bit-level programming yield. As shown in Fig. 4(b), the programmed state of each RRAM cell is verified after the write operation, and additional write pulses are applied only to cells that do not satisfy the verification condition. For cells that already satisfy the verification condition, the programming voltage is disconnected to reduce redundant programming attempts and unnecessary write-bias application.

3. Proposed RRAM macro design

3.1. Overall architecture

Fig. 5 (left) illustrates the overall architecture of the 1 Mb RRAM macro, which consists of two 512 Kb sub-macros, an error checking code (ECC) encoder and a decoder. Each 512 Kb sub-macro consists of an RRAM subarray, a row decoder, a column decoder, WL/SL/BL drivers, 72× voltage-mode sense amplifiers (VSAs), and write verification circuits. The proposed 1 Mb RRAM macro includes a reference array and a write-verification block, compared with the prior 1 Mb macro. The RRAM macro also activates only one sub-macro for each operation to reduce unnecessary power consumption. Each RRAM sub-array consists of 512×1152 RRAM cells, with 18 columns for generating the reference voltages, and 128 columns for storing parity bits. In the prior RRAM macro, two replica columns are shared by 72 SAs. However, to reduce reference variation, 72 SAs share 18 replica columns, which are evenly distributed across the RRAM array (i.e., every two replica columns are located at the ends of the 128 RRAM columns). Fig. 5(right) demonstrates the diagram between the RRAM array and SAs. A 16-column RRAM array connects to the SA input via a 16-to-1 multiplexer. Replica cells are shared among 8 SAs to generate the reference voltage. When writing data, a 16-to-1 multiplexer connects the BL and SL, allowing 72 data bits to be written simultaneously.

3.2. Proposed bipolar forming driver

Fig. 6 shows the proposed BL and SL driver structures. The BL and SL

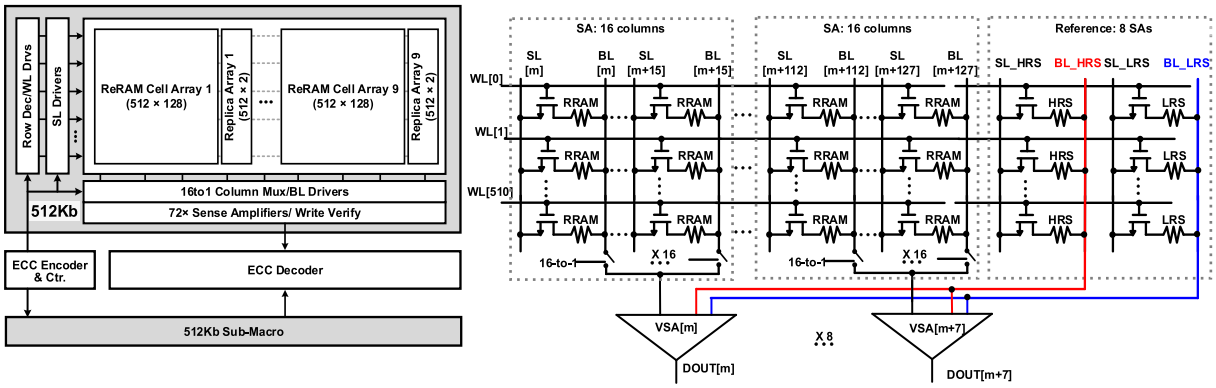


Fig. 5. Overall architecture of the 1 Mb RRAM macro.

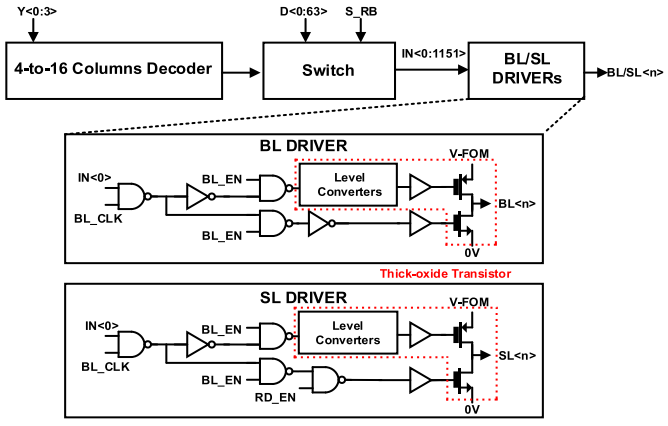


Fig. 6. Proposed BL/SL driver.

drivers are implemented as separate terminal-specific driver circuits because the BL and SL terminals require different bias polarities and control paths. However, additional dedicated forming-only driver circuits are not required. The same BL and SL drivers are reused for forming, SET, RESET, and read operations depending on the operation mode and the Form_EN signal. The BL/SL drivers are controlled by BL_CLK and BL_EN. The switch block signals, D < 0:63> and S_RB, determine whether the forming voltage is applied to the BL or SL side. For example, when both D < 0:63> and S_RB are high, the forming voltage is driven to the SL. Conversely, when both D < 0:63> and S_RB are low, the forming voltage is driven to the BL. These signal operations support both unipolar and bipolar forming operations. Although the use of separate BL and SL driver paths introduces some hardware overhead, the area impact is limited because the BL/SL driver circuits occupy only a small portion of the macro peripheral circuitry and are shared across forming, SET, RESET, and read operations.

Fig. 7 shows the waveforms for unipolar and bipolar forming operations. WL_SEL is used as a control signal to synchronize and enable the selected WL driver. When WL_SEL is activated, the selected WL<n> is driven to V_{FOM}, turning on the access transistor of the selected 1T1R cell. In unipolar forming, WL<n> is activated together with SL<n>, while BL<n> is held at 0 V. In bipolar forming, WL<n> is synchronized with the alternating BL/SL forming pulses so that the intended bipolar forming bias is applied to the selected cell. Unlike unipolar forming, where the forming bias is applied through the SL path, bipolar forming uses both the SL and BL paths alternately for the forming operation.

3.3. Proposed write verification

The basic concept of the write verification scheme is to compare RRAM data with external input data to determine whether to generate a

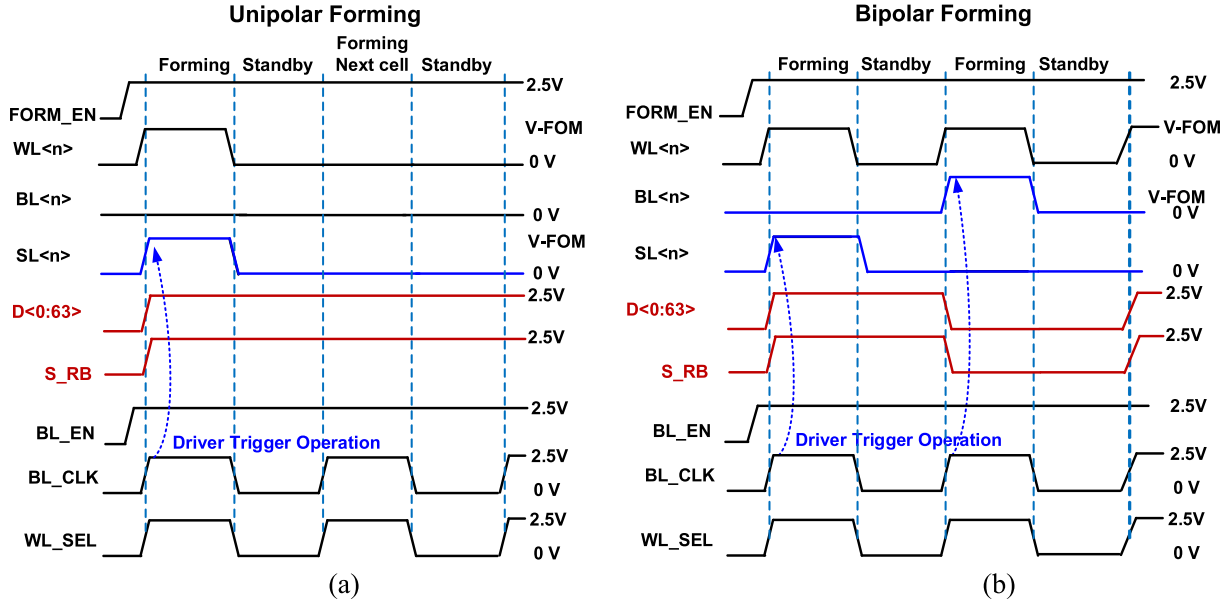


Fig. 7. Timing waveform of (a) unipolar forming, and (b) bipolar forming.

write voltage pulse. Fig. 8 shows the flowchart. The write verification operation is divided into three steps: before the write verification step, the initial states are read, and the results are compared with the input data to decide whether to proceed with write verification. During write verification operations, writing, reading, and comparison are repeated until the input data (DI) and output data (DO) match. Fig. 9 illustrates the detailed write-verification structure, which consists of 72 XNOR gates and 72 NAND gates. The XNOR stage compares the input data DI with the read data DO from the sense amplifier. If DI and DO are identical, the comparison result indicates that the target state has been reached. This comparison result is combined with the WR_VER signal through the NAND stage to generate the verification control signal for the following MUX.

The MUX controls whether the programming bias is delivered to the BL/SL drivers. If the verification condition is not satisfied, the MUX enables the corresponding BL or SL driver path so that the selected cell receives the required SET or RESET bias. If the verification condition is satisfied, the MUX disables the BL/SL programming paths, and no additional programming voltage is applied to the verified cell. The output is split into BL and SL paths because SET and RESET operations require different bias polarities. During SET, the SL path is driven while the BL path is held at 0 V. During RESET, the BL path is driven while the SL path is held at 0 V. Since the proposed macro adopts a split-SL structure, this separated BL/SL control enables cell-by-cell write verification across 72 cells. Therefore, cells that already satisfy the verification condition are protected from redundant programming, while cells that do not satisfy the condition continue the write operation.

4. Simulation

Fig. 10 shows the transient BL and SL voltage waveforms for bipolar forming, SET, and RESET operations. The transient simulation was performed to verify the circuit-level timing and bias-generation operation of the proposed BL/SL driver. For the simulation, a calibrated behavioral 1T1R HfOx RRAM model was adopted from the previously reported RRAM device/circuit co-design and sensing simulation frameworks [29,30]. The RRAM model uses two resistance states, HRS and LRS, and the cell current is determined by the applied voltage across the RRAM device and the corresponding resistance state, *i.e.*, $I_{RRAM} = V_{RRAM}/R_{STATE}$. The nominal resistance values were calibrated from

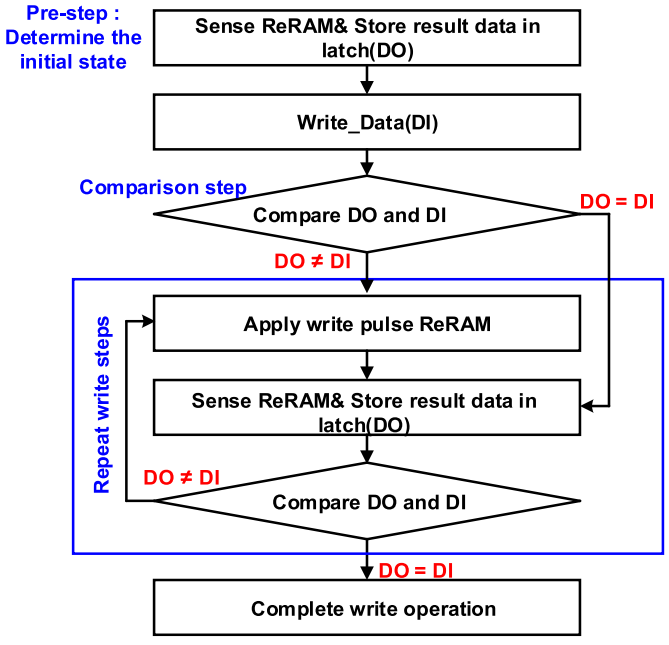
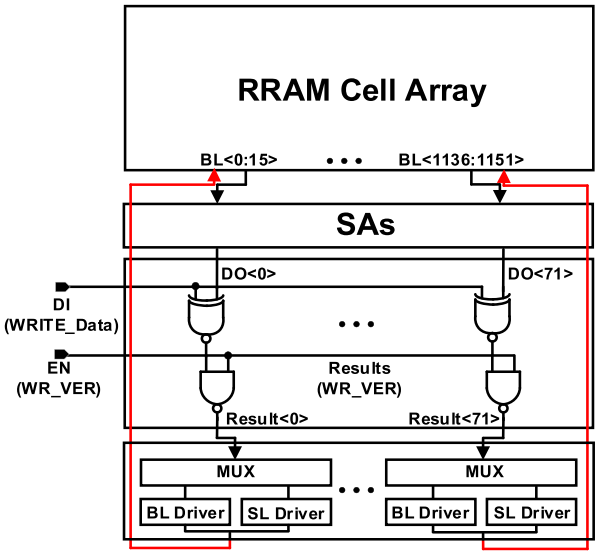


Fig. 8. Flowchart of write verification.



Comparison (DO and DI)	EN (WR_VER)	Result (WR_VER)	SL	BL
0(DO≠DI)	1	1	V_{SL-SET}	V_{SL-RST}
1(DO=DI)	1	0	0	0

Fig. 9. Write verification scheme and truth table.

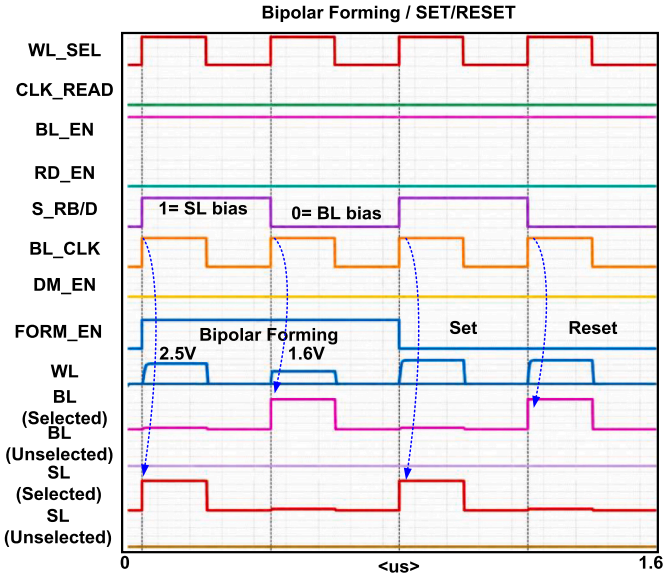


Fig. 10. Simulation waveform of bipolar forming pulse and write pulse.

reported HfOx RRAM measurement data, with $R_{HRS} = 950\,k\Omega$ and $R_{LRS} = 9\,k\Omega$ used as representative resistance states. SET/forming and RESET operations were modeled as state transitions between HRS and LRS based on the applied terminal-bias switching conditions. For the forming simulation, the unipolar forming condition used WL = 2.5 V, whereas the bipolar forming condition used a lower WL voltage of WL = 1.6 V. The simulation clock period was set to 1 μs. This behavioral model was used to evaluate forming/write pulse generation and write-verification timing, rather than to predict the full physical conductive-filament dynamics.

As described in Fig. 6, the terminal-specific BL and SL drivers are

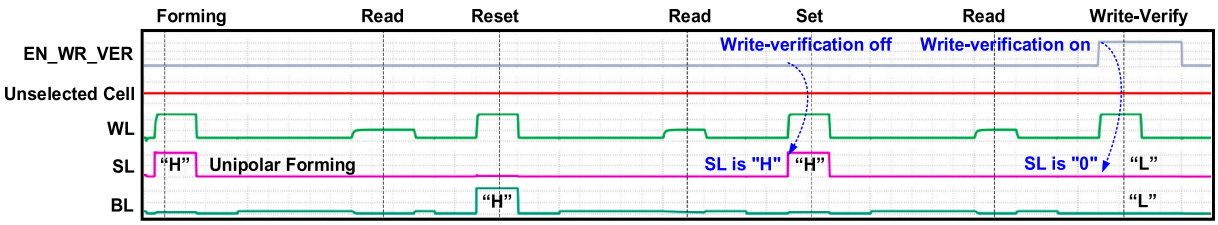


Fig. 11. Simulation waveform of programming and read including write verification.

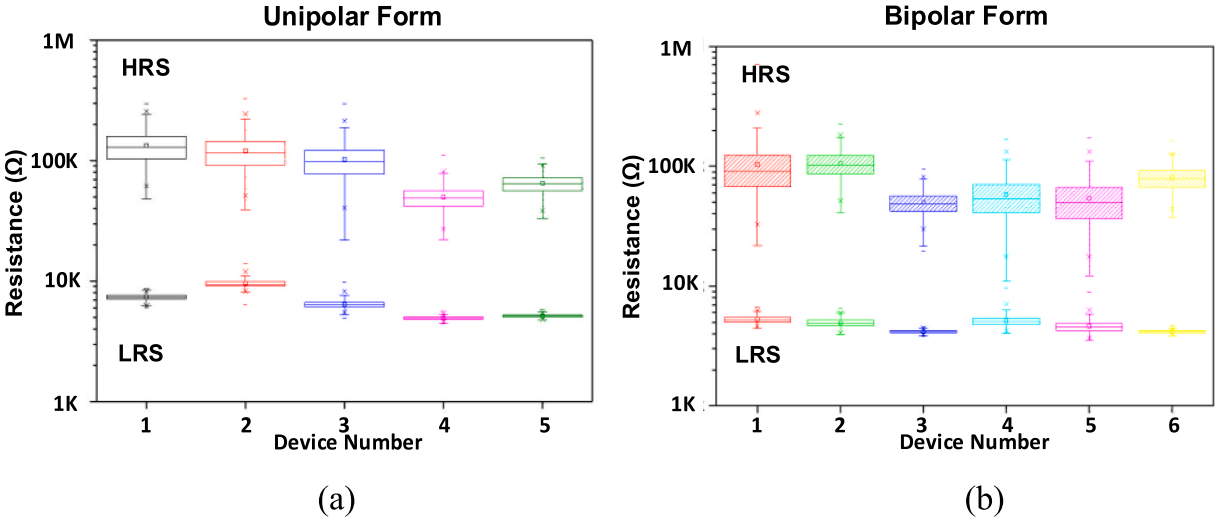


Fig. 12. Measurement result of (a) unipolar forming and (b) bipolar forming.

reused across forming, SET, RESET, and read operations, rather than using additional dedicated forming-only driver circuits. The simulation waveforms in Fig. 10 show that the reused BL/SL driver paths generate the bipolar forming pulse and the normal write pulse depending on the operation mode and the Form_EN signal. In addition, the write-verification waveform in Fig. 11 shows that the verification control signal determines whether the programming bias is delivered to the reused BL/SL driver paths or disconnected after successful verification. The operation shown in Fig. 10 is divided into two modes depending on Form_EN. When Form_EN is enabled, the write driver operates in the forming mode. In this mode, bipolar forming is controlled by Form_EN, S_RB, D, WL_SEL, and BL_CLK. According to the S_RB and D control signals, the forming voltage is selectively applied to the BL and SL paths, while the WL bias is applied at different levels during the forming sequence. As shown in Fig. 13(c), the implemented macro supports external bias ranges of 1.2–2.5 V for WL and 1.2–3.5 V for BL/SL. Therefore, during bipolar forming, the WL, BL, and SL bias levels can be independently controlled to evaluate a wide range of forming conditions. When Form_EN is disabled, the write driver operates in the normal programming mode and generates the corresponding SET or RESET bias conditions.

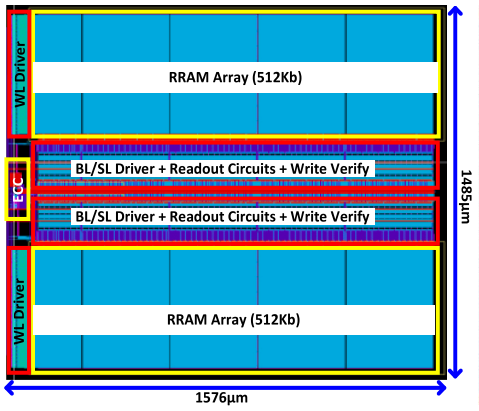
Fig. 11 shows the programming waveform including the write-verification operation. After unipolar forming, the proposed scheme performs the RESET, SET, and read verification operations step by step. In the first sequence, after the RESET operation, the following SET operation is normally performed, and the RRAM cell is programmed to the target SET state. Then, when the second SET operation is attempted with write verification, the verification result indicates that the cell has already completed the SET operation. Therefore, the SET programming path is disabled, and the SL voltage remains at 0 V during the SET period. As a result, no additional programming voltage is applied across the RRAM cell, preventing redundant programming once the target state is achieved.

5. Measurement result and analysis

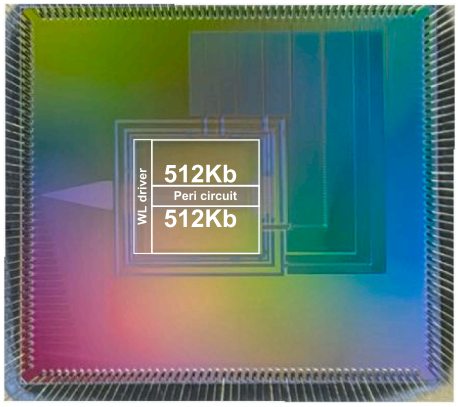
5.1. Measurement framework

Fig. 12 shows the preliminary measurement results of a single 1T1R RRAM device cell under unipolar and bipolar forming conditions. The unipolar forming operation was performed with a forming voltage of 2.5 V, whereas the bipolar forming operation was evaluated with forming voltages of 1.9 V and 1.8 V. These device-level results indicate that bipolar forming can reduce the required forming voltage compared with unipolar forming. Based on this observation, the proposed 1-Mb RRAM macro incorporates a write driver that supports both unipolar and bipolar forming operations, which are further evaluated through macro-level silicon measurements.

Fig. 13 shows the layout, die photograph, and summary table of the proposed 1 Mb RRAM macro. The macro consists of two 512 Kb sub-macros fabricated using a 40 nm CMOS process. It incorporates a write driver that supports both unipolar and bipolar forming operations, as well as a write-verification scheme for improved programming yield. Fig. 14 presents (a) the structure of the test chip and (b) the captured read waveform in test mode. After programming, the data stored in the RRAM array is read out through the DOUT_IO pin to verify the programming result. Based on this readout result, the programming yield is evaluated. In this work, the switching rate is used as a general term for the successful state-transition ratio of RRAM cells. For the forming operation, the switching rate is defined as the forming yield, which represents the ratio of fresh cells that are successfully formed. For SET/RESET programming, it is defined as the programming yield, which represents the ratio of cells that successfully switch to the target resistance state. Fig. 14(c) shows the programming test flow used to evaluate the proposed write-verification scheme. In the pre-step, the sense amplifier operation is verified by adjusting the reference voltage before programming. After each programming operation, the data stored in the



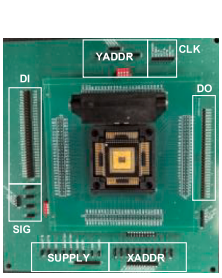
(a)

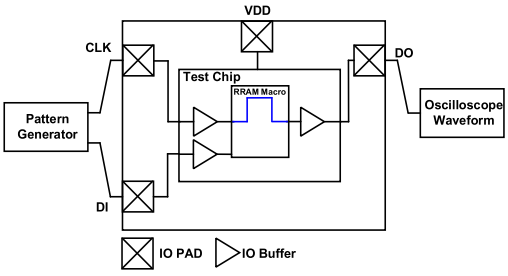


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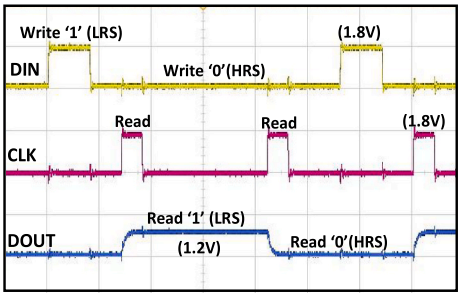
Technology	40nm Process
Capacity	1Mb (2 × 512Kb)
Architecture	Split SL
Sub-Array (512Kb)	512 × 1024
Core Area	1576μm × 1485μm
Cell Size	0.41μm × 0.328μm
Key Feature	1: Unipolar / Bipolar Forming 2: Write Verification
Amplitude	WL:1.2V- 2.5V BL:1.2V-3.5V SL:1.2V-3.5V

Fig. 13. (a) Layout of the 1 Mb RRAM macro and (b) Die photo with Summary table.

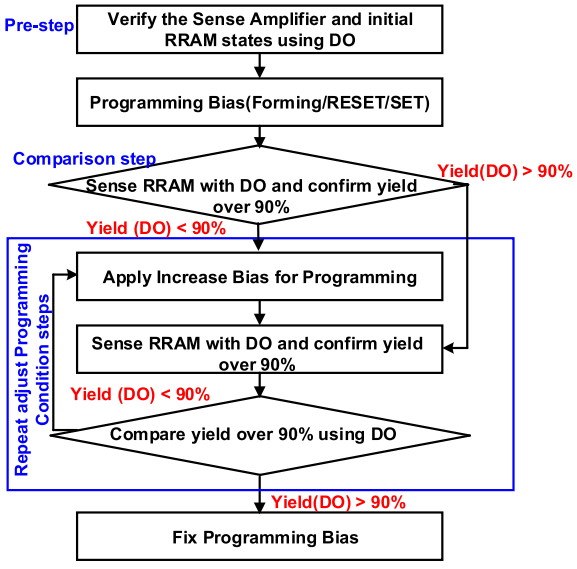




(a)



(b)



(c)

Fig. 14. (a) Setup and structure, (b) Measured read waveform, (c) Test flow chart.

RRAM array are read out through the DOUT_IO pin, and the programming result is verified at the bit level. Based on this readout result, the programming yield is evaluated.

The programming-yield test was performed using a small-array test setup. In this setup, 4096 bit cells were randomly selected across 10 dies to evaluate the bit-level programming-yield trend. Therefore, the measured result is presented as a sampled bit-level programming yield. The number of programming cycles ranged from 10 to 28 depending on the test condition. The forming and programming conditions were defined based on the externally applied WL/BL/SL terminal biases. Forming was performed with WL = 2.5 V, SL = 2.5 V, and BL = 0 V. SET was performed with WL = 2.0 V, SL = 2.0 V, and BL = 0 V, whereas

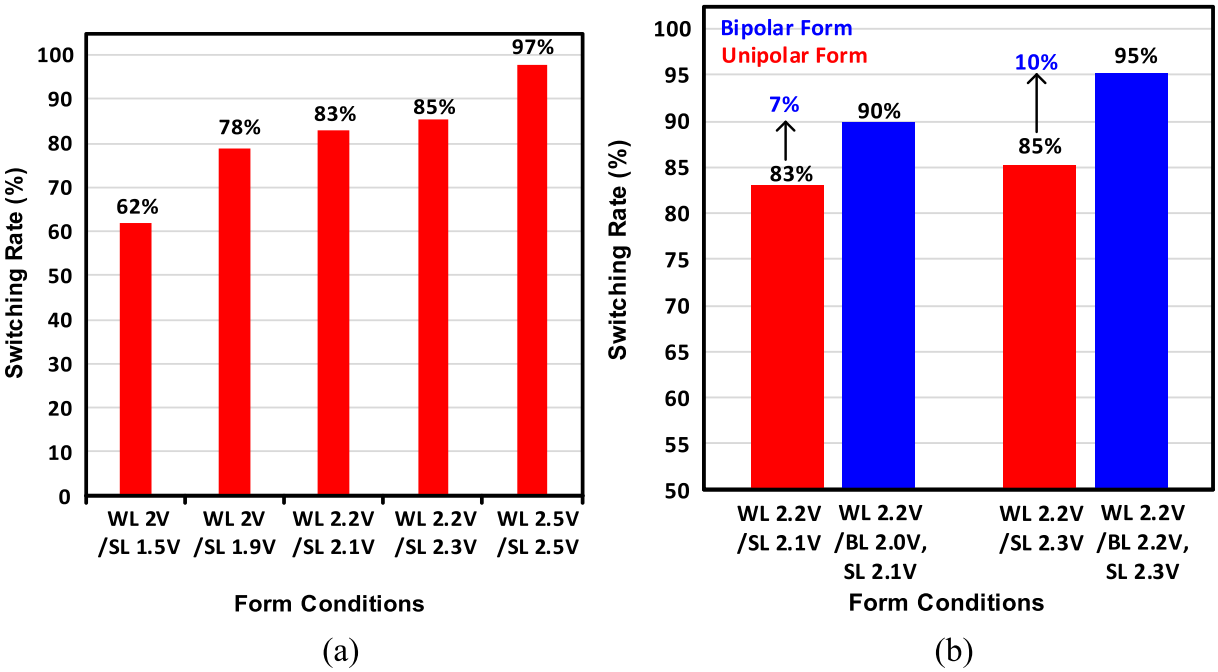


Fig. 15. (a) Measurement unipolar form yield with differents conditions and (b) Comparison of form yield between unipolar and bipolar.

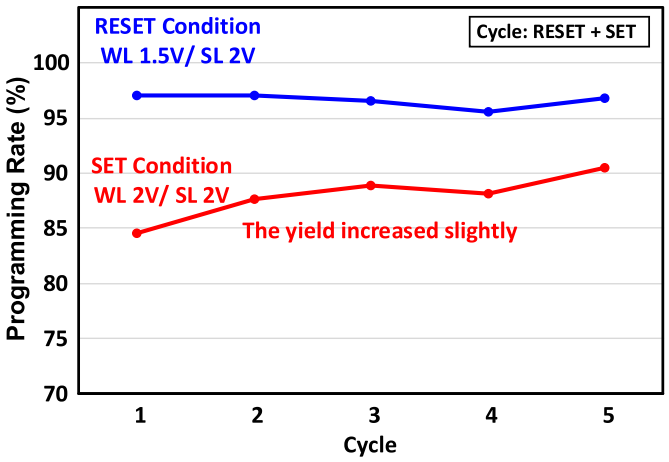


Fig. 16. Measurement Programming rate with cycle.

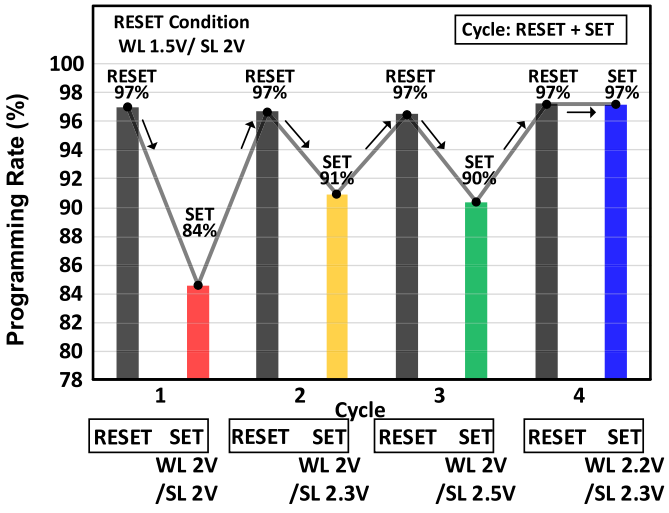


Fig. 17. Programming rate of SET based on different conditions.

RESET was performed with WL = 1.5 V, BL = 2.0 V, and SL = 0 V. The unselected WL, BL, and SL lines were held at 0 V. The pulse width was 1 μs, and the current compliance was set to 1 mA. Since the actual voltage across the RRAM stack was not directly measured in the test setup, the reported forming voltage refers to the externally applied terminal bias condition.

After the pre-step verification, the programming operation is performed, and the yield is evaluated through the DOUT_IO-based write-verification flow. If the yield is below 90%, the applied programming bias is incrementally increased until the yield exceeds 90%. Once the yield exceeds 90%, the corresponding bias condition is fixed and used for subsequent operations. Therefore, Fig. 14(c) presents the measurement framework for determining the programming-bias condition using the 90% yield criterion.

5.2. *Measurement result for bipolar forming*

Fig. 15 (a) presents the forming switching rate, where fresh HRS transitions to LRS through a unipolar forming process under five

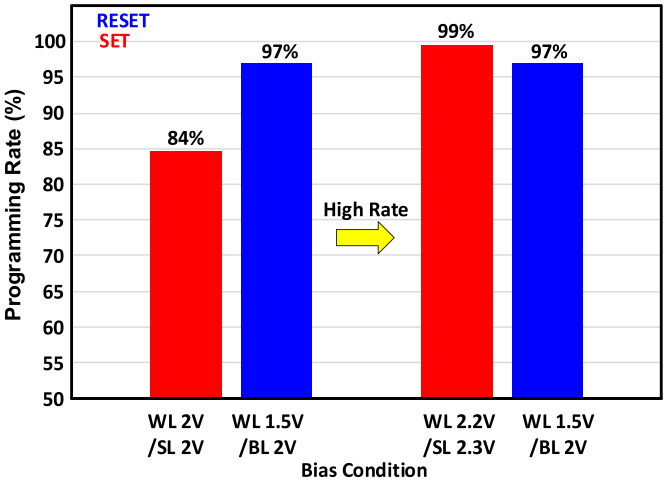


Fig. 18. Measuremenrt Programming rate under fixed condition.

different bias conditions. In this test, the switching rate was evaluated for each condition by varying WL and SL biases to investigate the sensitivity of the forming process to bias levels. As shown in Fig. 12, the target WL and SL voltages for the unipolar forming condition are 2.5V, which were previously optimized based on preliminary silicon data. Under this condition, the forming operation achieves a switching yield exceeding 90% in the fabricated 1 Mb RRAM macro, as shown in Fig. 15 (a), demonstrating its high forming reliability. However, to further analyze the forming behavior, Fig. 15(b) compares the switching yields of the unipolar and bipolar forming modes under identical bias conditions. In the case of unipolar forming, the switching rate exceeds approximately 80% under the third (WL = 2.5 V, SL = 2.1 V) and fourth (WL = 2.5 V, SL = 2.3 V) conditions. Based on these observations, the bipolar forming operation is subsequently performed under the same bias conditions to enable a direct comparison with the unipolar forming case. As a result, bipolar forming achieved a switching rate exceeding 90% by improving the yield 7% and 10% under each condition. These results clearly demonstrate the advantage of bipolar forming over unipolar forming. Furthermore, bipolar forming in the fourth condition achieved a similar switching yield at a lower voltage compared to unipolar forming of the fifth condition (WL = 2.5V, SL = 2.5V). Therefore, these results indicated that bipolar forming provides a more effective scheme for RRAM macro forming.

5.3. *Write verification programming and analysis*

The programming yield was evaluated based on the read data obtained through the DOUT_IO path after each programming operation. A SET operation was regarded as successful when the programmed cell was read as the target LRS data, whereas a RESET operation was regarded as successful when the programmed cell was read as the target HRS data. In this work, one programming cycle denotes one attempted SET or RESET operation followed by a verify-read operation. Fig. 16 shows the RESET and SET programming yields after forming as a function of repeated programming cycles. Under the initial target condition, the RESET operation achieves a high yield in the first programming cycle, whereas the SET operation yields only 84%. Therefore, multiple SET cycles were applied to evaluate the effect of the write-verification scheme. Although repeated SET operations were successfully performed, the overall yield improvement was limited, reaching an average value of approximately 80%. This result indicates that repeated programming with write verification can partially improve the programming result, but it is not sufficient to achieve robust yield improvement under a non-optimized bias condition.

Fig. 17 shows the SET programming yield under different bias conditions. The programming yield was evaluated by repeating RESET operations with a fixed bias and applying SET cycles under varying SET-bias conditions. Among the evaluated conditions, Condition 4 (WL = 2.2 V, SL = 2.3 V) achieved a SET programming yield of 97%. In addition, after 10 repeated cycles, an average yield of 97% was obtained. These results demonstrate that the programming bias condition has a stronger impact on yield improvement than simply increasing the number of programming cycles. Due to intrinsic RRAM variation, the yield can also differ among arrays even under the same bias condition; nevertheless, the same condition showed an average yield of 98% in another array macro.

Fig. 18 compares the programming yield under the initial target condition and the adjusted bias condition. The comparison confirms that the measured yield can be improved by selecting an appropriate programming bias based on the test flow. This result directly supports the need for a measurement-guided bias determination method rather than relying only on repeated write-verification cycles.

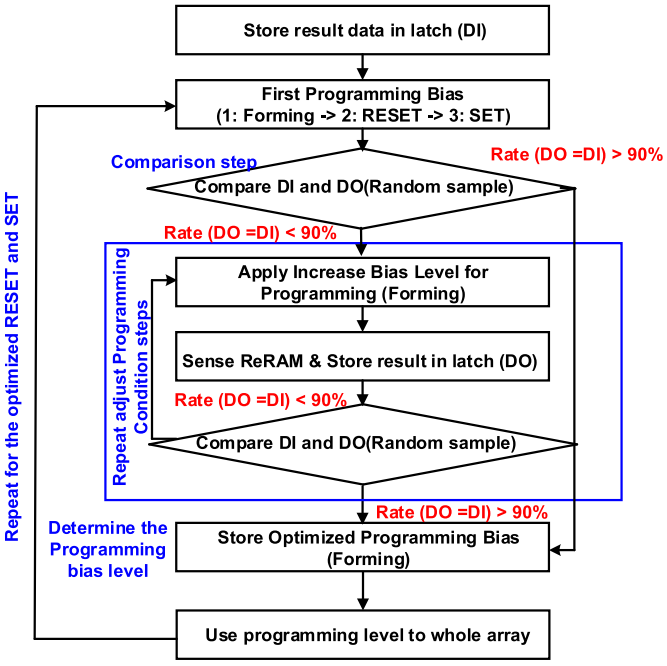


Fig. 19. Pre-step programming bias determination for high programming yield.

Fig. 19 shows the flow chart of the proposed programming-bias determination method. Before programming the entire array, the full programming sequence, including FORMING, RESET, and SET operations, is performed on randomly selected sample cells. From these measurements, the programming bias condition that satisfies the 90% yield criterion is identified and stored. The selected condition is then used for subsequent array programming. This method uses the write-verification result to determine a yield-robust programming condition and reduces redundant programming attempts compared with simply increasing the number of programming cycles.

To compare the proposed macro with representative prior RRAM forming, write-control, and write-verification works, Table 1 summarizes key design features and measurement results. Previous forming-related studies have mainly focused on reducing the forming voltage at the device level, while prior write-verification and write-control schemes have targeted multilevel conductance tuning, resistance-threshold-based self-write termination, sneaking-current suppression, or adaptive write termination. In addition, macro-level adaptive forming/programming schemes have been reported using adaptive forming/set/reset operation, pulse-width/voltage/current write-verify-write sequences, or self-adaptive write/read assist circuits.

In contrast, this work does not claim novelty solely based on macro-level adaptive operation. Instead, this work demonstrates a fabricated 1-Mb RRAM macro that supports both unipolar and bipolar forming operations through a dual-mode forming driver. The dual-mode forming operation is integrated with cell-by-cell write verification, enabling sampled bit-level programming-yield evaluation. Therefore, this work focuses on macro-level silicon validation of dual-mode unipolar/bipolar forming support, its integration with cell-by-cell write verification, and measurement-guided programming-bias selection based on sampled bit-level yield evaluation.

Table 1
Comparison with state-of-the-art FORMING and write-verification WORKS.

	Xue et al., JSSC'2013 [24]	Jain et al., ISSCC'2019 [8]	Zheng et al., EDL'2018 [21]	Shim et al., SST'2020 [28]	Chen et al., IEDM'2017 [23]	Yang et al., VLSI'2020 [31]	Yang et al., ISSCC'2021 [32]	This work
Technology/ Capacity	0.13 μ m logic based /8 Mb	22 nm FinFET/ 3.6 Mb	40 nm/Single 1T1R device	90 nm/64 kb	0.15 μm/16 Mb	28 nm/1.5 Mb	14 nm FinFET/1 Mb	40 nm/1 Mb
Supply or Bias Range	1.2V/3.3V	0.8V	N/R	N/R	1.8 V	1.8 V I/O, 1.05 V core	0.85 V	2.5 V I/O 1.2 V core
Silicon Measurement Scope	SAWM	TFR-based 1T1R	Device-level 1T1R measurement	Small-array/cell-array level	Macro-level silicon with sampled-cell measurement	Macro-level silicon	Macro-level silicon	Macro-level silicon with bit-level small-array yield measurement
Application	Write verification	Write verification	Bipolar Forming	Write verification	Write verification	Write verification	Write verification	Unipolar/Bipolar Forming, Write verification
Main Scheme	SAWM	PVC-WVW sequence	Bipolar ISPP	Two-step write–verify	DV-SWT self-write termination	1T2R + self-adaptive write driver	SADT + RESET write–verify	Dual-mode forming driver supporting unipolar/bipolar forming + cell-by-cell write verification
Forming Support Voltage	$Vw_{rite} = 3.3V$	N/R	Gate = 0.85 V, TE ISPP = 1.5–3.5V, Source = 0–0.75 V	N/R	N/R	N/R	SL = 0.6-1V, BL = −0.6V	WL = 1.2- 2.5 V, SL = 1.2-3.5 V, BL = 1.2-3.5V
Write Pulse Condition	N/R	Write pulse <10us	Forming pulse = 1us	SET/RESET pulse 100-300ns	N/R	N/R	N/R	Write pulse = 1us
Verification Scope	SET/RESET verification	SET/RESET verification	Device level Forming voltage measurement	Conductance target range	Resistance switching threshold	Current-limited SET/RESET write–verify	Cell-adaptive SET termination/RESET write–verify	Forming yield, programming-yield verification
Key Result	Reset 100%	10.1 Mb/mm^2 ,Write time <10us	Forming voltage reduction by 10%	Fine-tuning target	Tail current reduction	Sneaking current reduction >90%	RON-related failures reduced by up to 87%	Bipolar forming-yield improvement of 7–10% compared with unipolar forming , programming yield >90%; SET yield up to 97%

N/R: not reported.

6. Conclusion

In this paper, a new 1 Mb RRAM macro consisting of two 512 Kb sub-macros is proposed. it is fabricated using a 40 nm process with logic-compatible RRAM devices. The proposed macro incorporates two key design features. First, a write driver supporting bipolar forming operation is implemented to reduce the forming voltage. Second, a write-verification scheme with a split source line (SL) structure is adopted to improve write yield compared with the previous 1 Mb macro [19]. Measurement results demonstrate that the bipolar forming implemented in the write driver improves yield even when the same forming voltage is applied across the macro. The proposed forming method achieves a forming yield comparable to that of unipolar forming, which requires a higher forming voltage. It also shows that multiple programming can be effectively performed to enhance programming yield. Furthermore, the programming bias condition has a greater impact on yield than multiple programming. Therefore, new pre-step programming techniques are proposed to further improve programming yield for the next silicon implementation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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